

Task 1: Install and Authenticate

Install the GitHub CLI on your machine

Authenticate with your GitHub account

Verify you're logged in and check which account is active

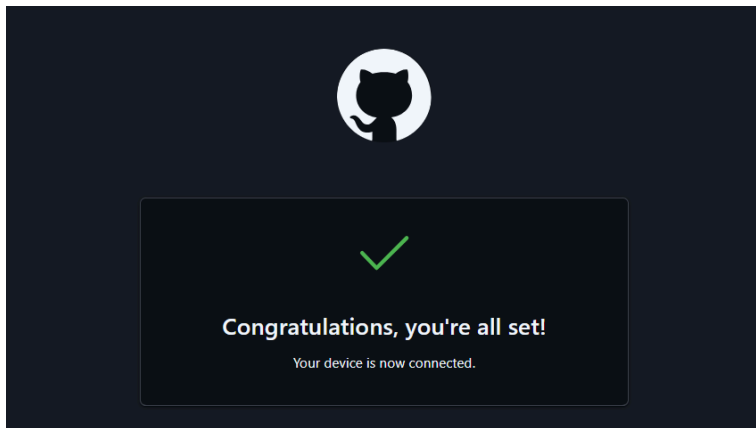
```
PS C:\Users\allen> winget install GitHub.cli
Found GitHub CLI [GitHub.cli] Version 2.88.1
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/cli/cli/releases/download/v2.88.1/gh_2.88.1_windows_amd64.msi
13.6 MB / 13.6 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\Users\allen> |
```

```
PS C:\Users\allen> gh --version
gh version 2.88.1 (2026-03-12)
https://github.com/cli/cli/releases/tag/v2.88.1
PS C:\Users\allen> |
```

```
PS C:\Users\allen> gh auth login
? Where do you use GitHub? GitHub.com
? What is your preferred protocol for Git operations on this host? SSH
? Upload your SSH public key to your GitHub account? Skip
? How would you like to authenticate GitHub CLI? Login with a web browser

! First copy your one-time code: 39DC-3135
Press Enter to open https://github.com/login/device in your browser... |
```

```
Press Enter to open https://github.com/login/device in your browser...
✓ Authentication complete.
- gh config set -h github.com git_protocol ssh
✓ Configured git protocol
✓ Logged in as samsrajyt
PS C:\Users\allen> |
```



```
PS C:\Users\allen> gh auth status
github.com
✓ Logged in to github.com account samsrajyt (keyring)
- Active account: true
- Git operations protocol: ssh
- Token: gho_*****
- Token scopes: 'gist', 'read:org', 'repo'
PS C:\Users\allen>
```

Answer in your notes: What authentication methods does gh support?

- Browser-based Auth
- Personal Access Token (PAT)
- SSH Key-based

Task 2: Working with Repositories

1. Create a new GitHub repo directly from the terminal — make it public with a README

```
? What would you like to do? Create a new repository on github.com from scratch
? Repository name gh-cli-day26-task

? Repository name gh-cli-day26-task
? Description GitHub CLI practice

? Description GitHub CLI practice
? Visibility Public
? Would you like to add a README file? Yes
? Would you like to add a .gitignore? Yes
? Choose a .gitignore template Actionscript
? Would you like to add a license? No
? This will create "gh-cli-day26-task" as a public repository on github.com. Continue? Yes
✓ Created repository samsrajyt/gh-cli-day26-task on github.com
https://github.com/samsrajyt/gh-cli-day26-task
? Clone the new repository locally? No
PS C:\Users\allen>
```

2.Clone a repo using gh instead of git clone

```
PS C:\Users\allen> gh repo clone samsrajyt/gh-cli-day26-task
Cloning into 'gh-cli-day26-task'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
PS C:\Users\allen>
```

3.View details of one of your repos from the terminal

```
PS C:\Users\allen> gh repo view samsrajyt/gh-cli-day26-task
samsrajyt/gh-cli-day26-task
GitHub CLI practice

  gh-cli-day26-task

  GitHub CLI practice

View this repository on GitHub: https://github.com/samsrajyt/gh-cli-day26-task
PS C:\Users\allen>
```

4.List all your repositories

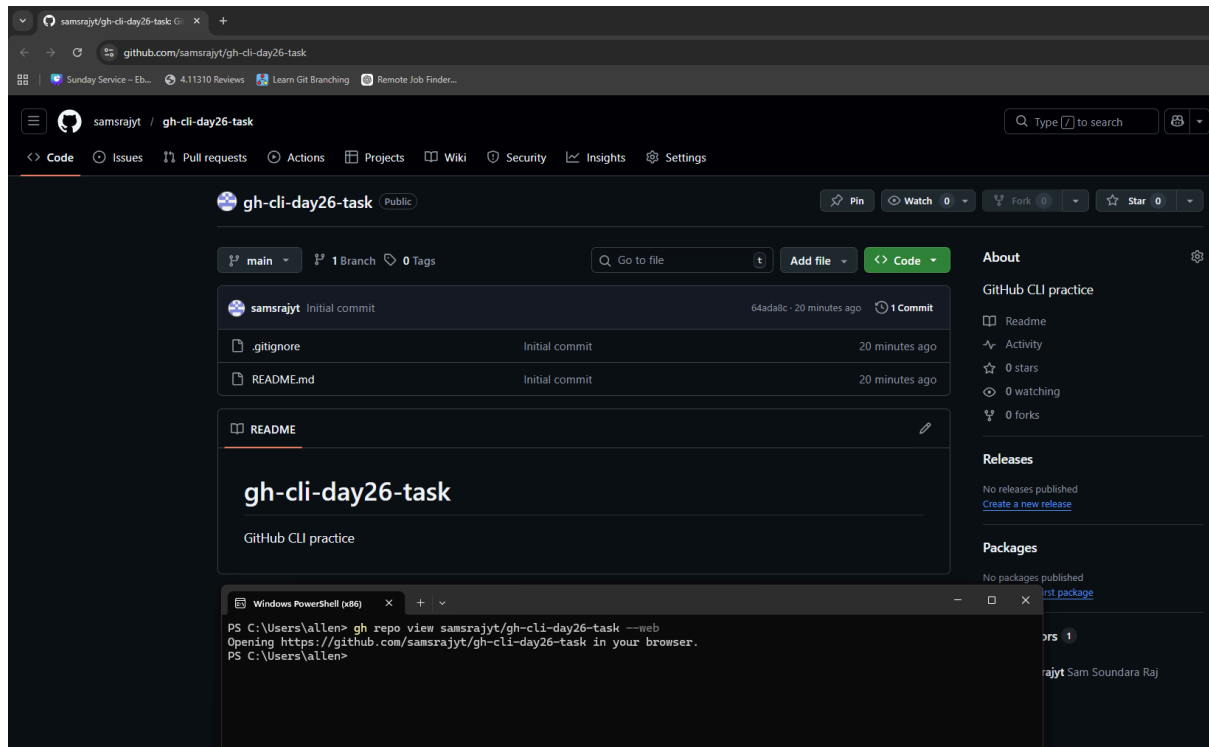
```
PS C:\Users\allen> gh repo list

Showing 12 of 12 repositories in @samsrajyt
```

NAME	DESCRIPTION	INFO	UPDATED
samsrajyt/gh-cli-day26-task	GitHub CLI practice	public	about 6 minutes ago
samsrajyt/90DaysOfDevOps	This repository is a Challenge for the DevOp...	public, fork	about 13 hours ago
samsrajyt/kubestarter	This is a Repository to get you started with...	public, fork	about 2 days ago
samsrajyt/devops-git-practice-remote	This is a repository for practice purpose	public	about 3 days ago
samsrajyt/rajesh-portfolio		public, fork	about 8 days ago
samsrajyt/devops-git-practice		public	about 28 days ago
samsrajyt/Shell-Scripting-For-DevOps		public, fork	about 28 days ago
samsrajyt/github-for-devops		public	about 28 days ago
samsrajyt/python-for-devops	Python For DevOps [AI Edition] is a hands-on...	public, fork	about 2 months ago
samsrajyt/tws-portfolio		public, fork	about 1 year ago
samsrajyt/GuviAssignments		public	about 2 years ago
samsrajyt/helloworld-java-maven		public, fork	about 2 years ago

```
PS C:\Users\allen>
```

5. Open a repo in your browser directly from the terminal



6. Delete the test repo you created (be careful!!)

```
PS C:\Users\allen> gh repo delete samsrajyt/gh-cli-day26-task
? Type samsrajyt/gh-cli-day26-task to confirm deletion: samsrajyt/gh-cli-day26-task
? Type samsrajyt/gh-cli-day26-task to confirm deletion: samsrajyt/gh-cli-day26-task
✓ Deleted repository samsrajyt/gh-cli-day26-task
PS C:\Users\allen>
```

Task 3: Issues

Create an issue on one of your repos from the terminal — give it a title, body, and a label

The screenshot shows a GitHub repository page for 'samsrajyt/github-for-devops' with the 'Issues' tab selected. The page displays a search bar with 'is:issue state:open', a filter for 'Open' issues (1), and a list of issues including a 'Feature bug' with the label 'enhancement'. Overlaid on this is a Windows PowerShell terminal window showing the following commands and output:

```
? Type samsrajyt/gh-cli-day26-task to confirm deletion: samsrajyt/gh-cli-day26-task
✓ Deleted repository samsrajyt/gh-cli-day26-task
PS C:\Users\allen> gh auth refresh -h github.com -s "repo,read:org,gist,workflow"

! First copy your one-time code: 6CF6-F33A
Press Enter to open https://github.com/login/device in your browser...
✓ Authentication complete.
PS C:\Users\allen> gh issue create --repo samsrajyt/github-for-devops

Creating issue in samsrajyt/github-for-devops

? Title (required) Feature bug
? Title (required) Feature bug
? Body <Received>
? What's next? Add metadata
? What would you like to add? Assignees, Labels, Projects, Milestone
? Assignees samsrajyt (Sam Soundara Raj)
? Labels enhancement
warning: no projects to choose from
warning: no milestones in the repository
? What's next? Submit
https://github.com/samsrajyt/github-for-devops/issues/2
PS C:\Users\allen>
```

List all open issues on that repo

```
PS C:\Users\allen> gh issue list --state open --repo samsrajyt/github-for-devops

Showing 1 of 1 open issue in samsrajyt/github-for-devops

ID  TITLE  LABELS  UPDATED
#2  Feature bug  enhancement  about 3 minutes ago
PS C:\Users\allen> |
```

View a specific issue by its number

```
PS C:\Users\allen> gh issue view 1 --repo samsrajyt/github-for-devops
Added Dev file samsrajyt/github-for-devops#1
Open • samsrajyt (Sam Soundara Raj) opened about 1 month ago • 0 comments

No description provided

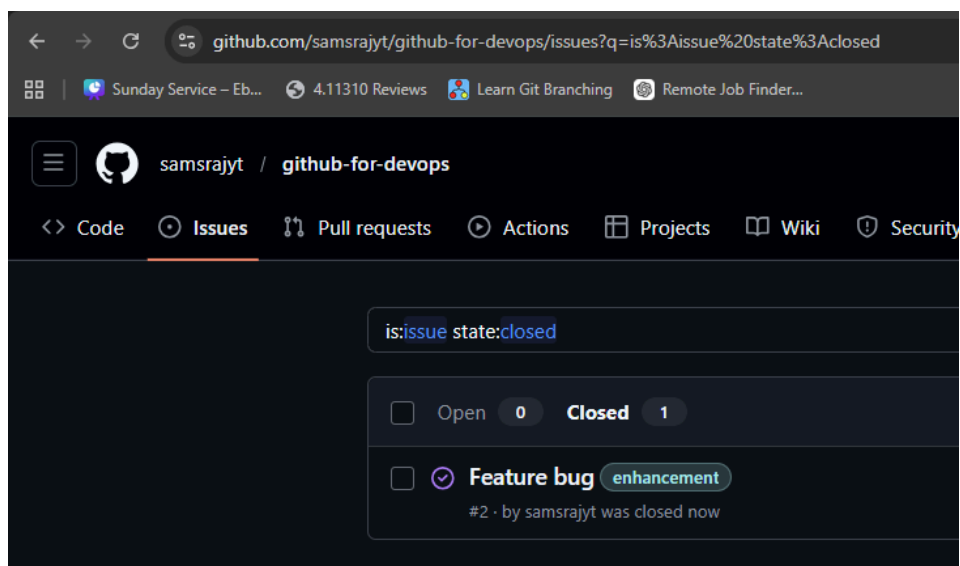
View this issue on GitHub: https://github.com/samsrajyt/github-for-devops/pull/1
PS C:\Users\allen> gh issue view 2 --repo samsrajyt/github-for-devops
Feature bug samsrajyt/github-for-devops#2
Open • samsrajyt (Sam Soundara Raj) opened about 7 minutes ago • 0 comments
Assignees: samsrajyt (Sam Soundara Raj)
Labels: enhancement

a new feature needs to be implemented for this project

View this issue on GitHub: https://github.com/samsrajyt/github-for-devops/issues/2
PS C:\Users\allen>
```

Close an issue from the terminal

```
PS C:\Users\allen> gh issue close 2 --repo samsrajyt/github-for-devops --comment "Issue Resolved"
✓ Closed issue samsrajyt/github-for-devops#2 (Feature bug)
PS C:\Users\allen>
```



How could you use gh issue in a script or automation?

We can create issues by integrating into a script and whenever the solution for a bug or feature is developed we can just close from the CLI.

Task 4: Pull Requests

Create a branch, make a change, push it, and create a pull request entirely from the terminal

```
allen@DESKTOP-CVA040U MINGW64 ~/github-for-devops (master)
$ git switch -c feature
Switched to a new branch 'feature'

allen@DESKTOP-CVA040U MINGW64 ~/github-for-devops (feature)
$ echo "Adding a feature" >> feature.html

allen@DESKTOP-CVA040U MINGW64 ~/github-for-devops (feature)
$ git status
On branch feature
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    feature.html

nothing added to commit but untracked files present (use "git add" to track)

allen@DESKTOP-CVA040U MINGW64 ~/github-for-devops (feature)
$ git add feature.html
warning: in the working copy of 'feature.html', LF will be replaced by CRLF the next time Git touches it

allen@DESKTOP-CVA040U MINGW64 ~/github-for-devops (feature)
$ git commit -m "Feature Added"
[feature 2203d52] Feature Added
 1 file changed, 1 insertion(+)
 create mode 100644 feature.html

allen@DESKTOP-CVA040U MINGW64 ~/github-for-devops (feature)
$ git remote -v
origin  https://github.com/samsraajt/github-for-devops.git (fetch)
origin  https://github.com/samsraajt/github-for-devops.git (push)

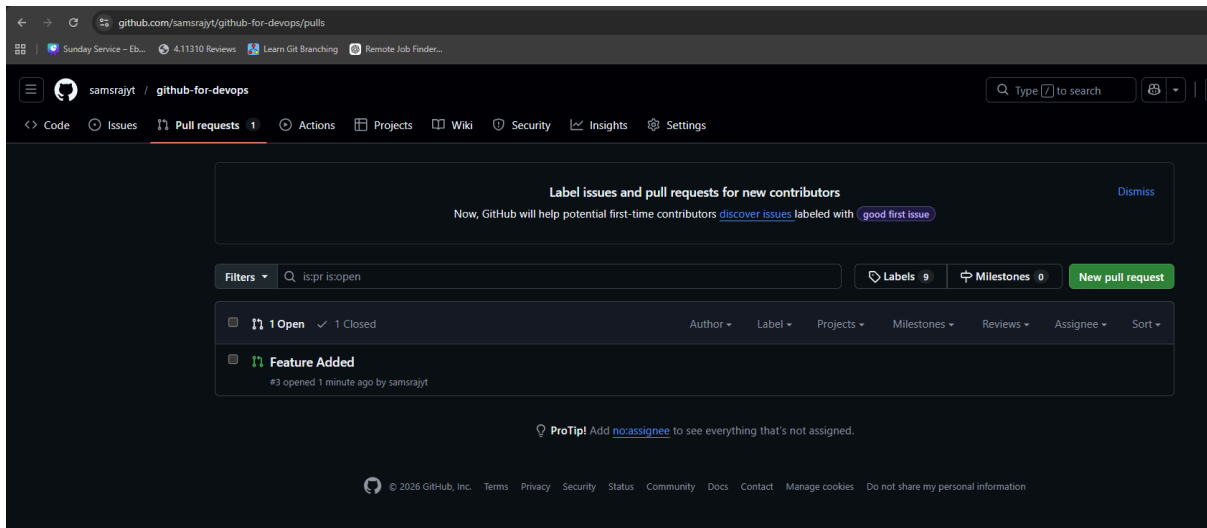
allen@DESKTOP-CVA040U MINGW64 ~/github-for-devops (feature)
$ git push origin feature
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 285 bytes | 285.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature' on GitHub by visiting:
remote:   https://github.com/samsraajt/github-for-devops/pull/new/feature
remote:
To https://github.com/samsraajt/github-for-devops.git
 * [new branch]      feature -> feature

allen@DESKTOP-CVA040U MINGW64 ~/github-for-devops (feature)
$ |
```

```
PS C:\Users\allen\github-for-devops\github-for-devops> gh pr create
head branch "master" is the same as base branch "master", cannot create a pull request
PS C:\Users\allen\github-for-devops\github-for-devops> git checkout feature
Switched to a new branch 'feature'
branch 'feature' set up to track 'origin/feature'.
PS C:\Users\allen\github-for-devops\github-for-devops> gh pr create

Creating pull request for feature into master in samsraajt/github-for-devops

? Title (required) Feature Added
? Body <Received>
? What's next? Submit
https://github.com/samsraajt/github-for-devops/pull/3
PS C:\Users\allen\github-for-devops\github-for-devops>
```



List all open PRs on a repo

View the details of your PR — check its status, reviewers, and checks

```
PS C:\Users\allen\github-for-devops\github-for-devops> gh pr list

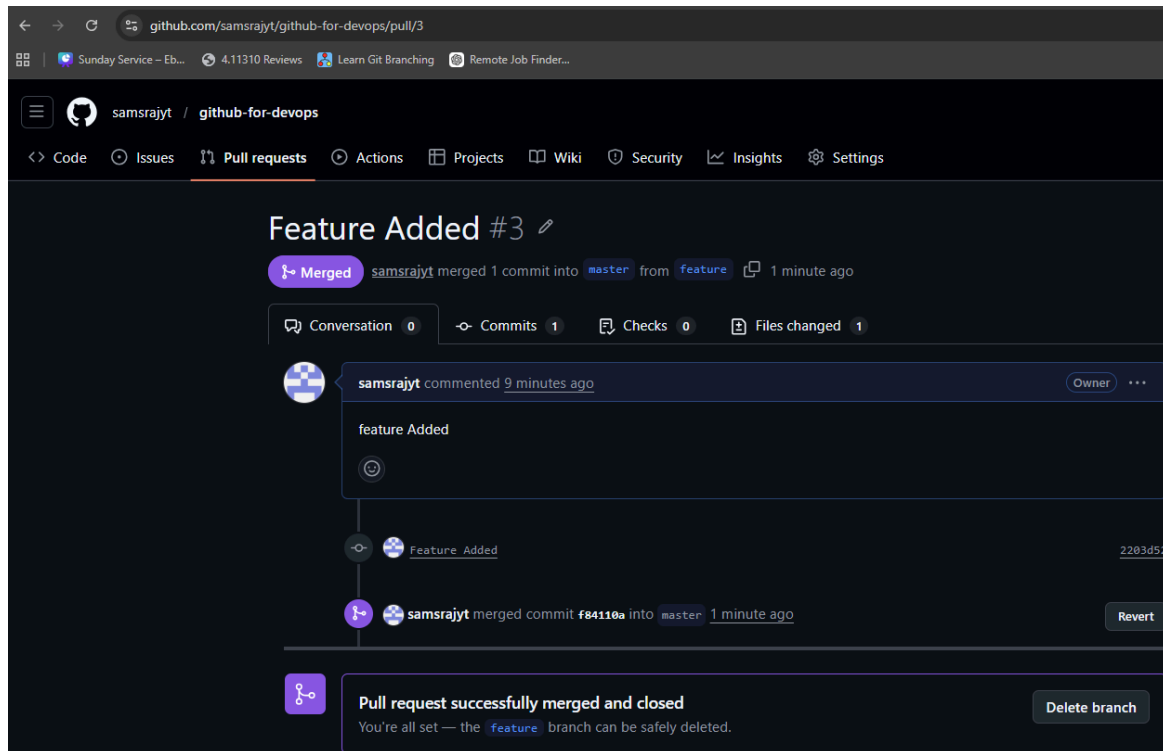
Showing 1 of 1 open pull request in samsrajyt/github-for-devops

ID   TITLE           BRANCH   CREATED AT
#3   Feature Added   feature  about 3 minutes ago
PS C:\Users\allen\github-for-devops\github-for-devops> |
```

```
PS C:\Users\allen\github-for-devops\github-for-devops> gh pr view 3 --json state,reviewRequests,statusCheckRollup
{
  "reviewRequests": [],
  "state": "OPEN",
  "statusCheckRollup": []
}
PS C:\Users\allen\github-for-devops\github-for-devops> |
```

```
PS C:\Users\allen\github-for-devops\github-for-devops> gh pr merge 3
Merging pull request samsrajyt/github-for-devops#3 (Feature Added)
? What merge method would you like to use? Create a merge commit
? Delete the branch locally and on GitHub? No
? What's next? Submit
✓ Merged pull request samsrajyt/github-for-devops#3 (Feature Added)
PS C:\Users\allen\github-for-devops\github-for-devops> |
```

Merge your PR from the terminal



What merge methods does gh pr merge support?

- Merge Commit
- Squash and Merge
- Rebase and Merge

How would you review someone else's PR using gh?

gh pr review <PR-id>

Task 5: GitHub Actions & Workflows (Preview)

List the workflow runs on any public repo that uses GitHub Actions

View the status of a specific workflow run

```
PS C:\Users\allen\github-for-devops\github-for-devops> gh run list --repo tensorflow/tensorflow

STATUS  TITLE                                WORKFLOW                                BRANCH  EVENT  ID              ELAPSED  AGE
-----  -
* PR #110592 CodeQL refs/pull/110592/head dynamic 23188952235 2m14s about 2 minutes ago
* PR #112520 CodeQL refs/pull/112520/head dynamic 23188782491 4m4s about 6 minutes ago
* PR #112534 CodeQL refs/pull/112534/head dynamic 23188678548 5m14s about 9 minutes ago
* Reverts 6d3c15970763290c1cbfc76f1a0aeab384b59bc5 Creates a GitHub Issue when a PR Rolled back via Commit to Master master push 23188640621 1s about 10 minutes ago
* Reverts 6d3c15970763290c1cbfc76f1a0aeab384b59bc5 Scorecards supply-chain security master push 23188640619 1m36s about 10 minutes ago
* Push on master CodeQL refs/pull/112534/head dynamic 23188639593 9m18s about 10 minutes ago
* PR #112515 CodeQL refs/pull/112515/head dynamic 23188639806 4m5s about 10 minutes ago
* PR #112533 CodeQL refs/pull/112533/head dynamic 23188618828 3m52s about 10 minutes ago
* PR #112532 CodeQL refs/pull/112532/head dynamic 23188584473 4m9s about 11 minutes ago
* PR #110592 CodeQL refs/pull/110592/head dynamic 23188487175 3m57s about 14 minutes ago
* PR #112497 CodeQL refs/pull/112497/head dynamic 23188439873 5m22s about 15 minutes ago
* PR #112446 CodeQL refs/pull/112446/head dynamic 23188411486 3m49s about 16 minutes ago
* PR #112532 CodeQL refs/pull/112532/head dynamic 23187979189 3m54s about 27 minutes ago
* PR #112453 CodeQL refs/pull/112453/head dynamic 23187882358 4m4s about 29 minutes ago
* PR #112399 CodeQL refs/pull/112399/head dynamic 23187814542 3m54s about 31 minutes ago
* PR #112020 CodeQL refs/pull/112020/head dynamic 23187784968 3m53s about 32 minutes ago
* PR #112515 CodeQL refs/pull/112515/head dynamic 23187737488 3m51s about 33 minutes ago
* PR #112500 CodeQL refs/pull/112500/head dynamic 23187587717 3m58s about 37 minutes ago
* PR #112430 CodeQL refs/pull/112430/head dynamic 23187538296 6m5s about 39 minutes ago
* Automated Code Change Creates a GitHub Issue when a PR Rolled back via Commit to Master master push 23187505688 2s about 39 minutes ago

PS C:\Users\allen\github-for-devops\github-for-devops>
```

```
PS C:\Users\allen\github-for-devops\github-for-devops> gh run view 23188952235 --repo tensorflow/tensorflow

✓ refs/pull/110592/head CodeQL · 23188952235
Triggered via dynamic about 5 minutes ago

JOBS
✓ Analyze (go) in 1m44s (ID 67379626918)
✓ Analyze (python) in 4m11s (ID 67379626922)
✓ Analyze (actions) in 1m27s (ID 67379626934)
```

Answer in your notes: How could `gh run` and `gh workflow` be useful in a CI/CD pipeline?

In a CI/CD pipeline, they are most useful for **triggering**, **monitoring**, and **debugging** builds without ever leaving your terminal.